

Atlas Labs HR Analytics - Complete Project Documentation

End-to-End Business Intelligence Solution: Power BI Dashboards, Data Modeling, DAX Measures & Workforce Insights

1. Project Overview & Business Objectives

The **Atlas Labs HR Analytics Project** is a comprehensive business intelligence solution designed to track, analyze, and optimize human resources operations. The core objective is to provide executive management with deep visibility into workforce dynamics, employee performance trends, demographic structures, and organizational attrition risks across the entire organization.

2. Detailed Dashboard Architecture (The 4 Core Views)

1. Executive Overview Dashboard

- Tracks high-level KPIs: Total Employees (1,470), Active Headcount (1,233), Inactive/Left (237).
- Displays overall Attrition Rate benchmark (16%).
- Visualizes historical hiring trends and active distribution across business units.

2. Demographics & Diversity Dashboard

- Analyzes workforce composition by Age groups, Gender diversity, and Marital Status.
- Evaluates ethnic background distribution and educational attainment levels.
- Identifies demographic concentrations across different organizational tiers.

3. Performance Tracker Dashboard

- Monitors annual employee performance ratings and historical review cycles.
- Categorizes staff into performance brackets (Needs Improvement, Meets Expectations, Exceeds, Above & Beyond).
- Correlates review scores with tenure and departmental output.

4. Attrition & Turnover Analysis Dashboard

- Deep-dive investigation into why and where employees leave Atlas Labs.
- Segments turnover by Department, Business Travel frequency, and Years at Company.
- Analyzes the direct impact of OverTime and compensation on resignation rates.

3. Data Modeling & Star Schema Architecture

The solution is built on a robust relational **Star Schema** inside Power BI to guarantee high query performance and seamless cross-filtering:

- **Fact Table:** **Fact Performance Rating** capturing granular employee performance scores, review dates, and evaluation metrics.
- **Dimension Tables:** **Dim Employee** (core demographic & profile attributes), **Dim Date**, **Dim Education Level**, and **Dim Satisfaction Level**.
- **Relationships:** Enforced One-to-Many (1:*) relationships with optimized single/bi-directional cross-filtering to prevent ambiguity.

4. Advanced DAX Measures

Key calculated measures engineered to power interactive visuals and dynamic KPI cards:

Total Employees Measure

TOTAL EMPLOYEE = DISTINCTCOUNT('Dim Employee'[EmployeeID])

Active Employees Measure

ACTIVE EMPLOYEE = CALCULATE([TOTAL EMPLOYEE], 'Dim Employee'[Attrition] = "No")

Inactive / Left Employees Measure

IN ACTIVE EMPLOYEE = CALCULATE([TOTAL EMPLOYEE], 'Dim Employee'[Attrition] = "Yes")

Overall Attrition Rate Measure

ATTRITION RATE = DIVIDE([IN ACTIVE EMPLOYEE], [TOTAL EMPLOYEE], 0)

Average Performance Rating Measure

Average Performance = AVERAGE('Fact Performance Rating'[PerformanceRating])

5. Strategic Insights & Management Recommendations

Analysis Focus	Key Finding & Root Cause	Strategic Management Recommendation
Departmental Turnover	Technology and Sales sectors exhibited disproportionately higher turnover rates compared to Operations.	Conduct salary benchmarking and introduce specialized retention incentives for technical talent.
OverTime Burnout	Employees subjected to frequent OverTime demonstrated a critical 31% attrition rate.	Re-evaluate resource allocation, workload distribution, and enforce work-life balance policies.
Tenure Vulnerability	Resignation risk heavily spikes during an employee's first two years at the company.	Upgrade the onboarding structure, expand mentorship programs, and implement 90-day check-ins.
Job Satisfaction	Strong statistical correlation identified between low satisfaction scores and voluntary exits.	Deploy regular pulse surveys and establish transparent feedback channels with department heads.